

Homework 3

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1 Introduction

In this write up we will be looking at the differences between application of on-policy and off-policy Monte Carlo Control (MCC) to the Farama Foundation Gymnasium *CartPole-v1* environment which consists of a pole hinged on a cart that can be moved left or right. The goal is to keep the pole vertical at each step and the agent is rewarded for every step the pole remains reasonably upright. The application of on-policy and off-policy MCC to this environment helps to illustrate the differences between these approaches particularly with regards to policy improvement and convergence.

2 On-Policy MCC

In on-policy MCC, the agent learns about a policy while following that same policy. The agent generates episodes by interacting with the environment using the current policy, and then updates the action-value function based on the returns received from the episodes experienced. For this implementation and the others in this write up we used python and the OpenAI Gymnasium library to create the environment and implement the MCC algorithms. The on-policy MCC algorithm can be summarized as follows:

1. **Algorithm parameter:** small $\epsilon > 0$

2. **Initialize:**

- $\pi \leftarrow$ an arbitrary ϵ -soft policy
- $Q(s, a) \in \mathbb{R}$ (arbitrarily), for all $s \in \mathcal{S}, a \in \mathcal{A}(s)$
- $Returns(s, a) \leftarrow$ empty list, for all $s \in \mathcal{S}, a \in \mathcal{A}(s)$

3. **Repeat forever (for each episode):**

(a) Generate an episode following π : $S_0, A_0, R_1, \dots, S_{T-1}, A_{T-1}, R_T$

(b) $G \leftarrow 0$

(c) **Loop for each step of episode, $t = T - 1, T - 2, \dots, 0$:**

i. $G \leftarrow \gamma G + R_{t+1}$

ii. **Unless** the pair S_t, A_t appears in $S_0, A_0, S_1, A_1 \dots, S_{t-1}, A_{t-1}$:

- Append G to $Returns(S_t, A_t)$
- $Q(S_t, A_t) \leftarrow \text{average}(Returns(S_t, A_t))$
- $A^* \leftarrow \arg \max_a Q(S_t, a)$ (with ties broken arbitrarily)
- For all $a \in \mathcal{A}(S_t)$:

$$\pi(a|S_t) \leftarrow \begin{cases} 1 - \epsilon + \epsilon/|\mathcal{A}(S_t)| & \text{if } a = A^* \\ \epsilon/|\mathcal{A}(S_t)| & \text{if } a \neq A^* \end{cases}$$

The python implementation can be found in the appendix as well as accompanying files. the primary hyperparameters in question for us are $\bar{Q}_0$ the optimistic initial action-value estimate, ϵ_a the initial exploration rate, and ϵ_b the terminal exploration rate.